

Abstract



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The "CRM XTART" project provides an integrated corporate software ecosystem engineered to streamline customer relationship management for Small and Medium-sized Enterprises (SMEs). This cross-disciplinary technical document details the seamless unification of five core computing fields inside the Higher Technical Degree of Multiplatform Applications Development (DAM). Structurally, the ecosystem comprises a highly normalized relational database implemented on MySQL fulfilling Third Normal Form (3FN) standards, a desktop enterprise software suite developed in Java SE utilizing the Data Access Object (DAO) pattern alongside a graphical user interface based on Swing components, and an executive administration dashboard rendered in semantic HTML5 and stylized dynamically via Tailwind CSS. Synchronous middleware and remote communication are facilitated through a Python RESTful API constructed with the Flask micro-framework. The entire distributed architecture is successfully deployed within a standalone virtual machine running Windows Server managed through XAMPP services, configuring firewall ingress restrictions, static IP mapping, and implementing backup solutions based on automated binary byte-stream object serialization.